

(6.3.) 2. You are given that $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\}$ is an orthogonal basis for $\mathbb{R}^4$, where

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 1 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} -2 \\ 1 \\ -1 \\ 1 \end{bmatrix}, \mathbf{u}_3 = \begin{bmatrix} 1 \\ 1 \\ -2 \\ -1 \end{bmatrix}, \mathbf{u}_4 = \begin{bmatrix} -1 \\ 1 \\ 1 \\ -2 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 4 \\ 5 \\ -2 \\ 2 \end{bmatrix}. \text{ Write } \mathbf{v} \text{ as the sum of two vectors, one in } \text{Span}\{\mathbf{u}_1\} \text{ and the other in } \text{Span}\{\mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\}.$$

Sol. Denote the span of $\{\mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\}$ by W . Now consider $\mathbf{v} - \text{proj}_W \mathbf{v}$. You will easily see that the orthogonality of our basis gives $(\mathbf{v} - \text{proj}_W \mathbf{v}) \cdot \mathbf{u}_i = 0$ for each $i = 2, 3, 4$. This means that the remainder vector $\mathbf{v} - \text{proj}_W \mathbf{v}$ is orthogonal to W . Since $\mathbf{u}_1$ is orthogonal to W , the projection of $\mathbf{v}$ onto the vector $\mathbf{u}_1$ completely explains $\mathbf{v} - \text{proj}_W \mathbf{v}$, not falling short or being too long. Thus all you have to do is project $\mathbf{v}$ to $\text{Span}\{\mathbf{u}_1\}$ and $\text{Span}\{\mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\}$, because $\mathbf{v}$ breaks exactly into two pieces, $\text{proj}_W \mathbf{v}$ and $\text{proj}_{\mathbf{u}_1} \mathbf{v}$. Note the role of orthogonality here.

Using above,

$$\begin{aligned} \text{proj}_{\mathbf{u}_1} \mathbf{v} &= \frac{\mathbf{u}_1 \cdot \mathbf{v}}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 = 2\mathbf{u}_1 \\ \text{proj}_W \mathbf{v} &= \mathbf{v} - 2\mathbf{u}_1 = \begin{bmatrix} 2 \\ 1 \\ -4 \\ 0 \end{bmatrix} \end{aligned}$$

sum up as $\mathbf{v}$.

(6.2.) 10. Show that $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthogonal basis for $\mathbb{R}^3$. Then express $\mathbf{x}$ as a linear combination of the $\mathbf{u}$'s.

Sol. Show that $\mathbf{u}_i \cdot \mathbf{u}_j = 0$ for $i \neq j$ with $i, j = 1, 2, 3$ to prove the orthogonality. Then the orthogonality gives the linear independence of the set. Any linearly independent set of 3 $\mathbb{R}^3$ vectors is a basis of $\mathbb{R}^3$, so this finishes the first part.

To get the linear combination, again project $\mathbf{x}$ onto each components of the basis, and we know that the sum of these 3 projections is precisely $\mathbf{x}$ due to the orthogonality of the basis.

$$\mathbf{x} = \frac{\mathbf{u}_1 \cdot \mathbf{x}}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + \frac{\mathbf{u}_2 \cdot \mathbf{x}}{\mathbf{u}_2 \cdot \mathbf{u}_2} \mathbf{u}_2 + \frac{\mathbf{u}_3 \cdot \mathbf{x}}{\mathbf{u}_3 \cdot \mathbf{u}_3} \mathbf{u}_3.$$

- (6.1.) 36. Suppose $\mathbf{y}$ is orthogonal to $\mathbf{u}$ and $\mathbf{v}$. Show that $\mathbf{y}$ is orthogonal to every $\mathbf{w}$ in $\text{Span}\{\mathbf{u}, \mathbf{v}\}$.
[Hint: An arbitrary $\mathbf{w}$ in $\text{Span}\{\mathbf{u}, \mathbf{v}\}$ has the form $\mathbf{w} = c_1\mathbf{u} + c_2\mathbf{v}$. Show that $\mathbf{y}$ is orthogonal to such a vector $\mathbf{w}$.]

Sol. As in the hint, any vector $\mathbf{w}$ in the span of $\{\mathbf{u}, \mathbf{v}\}$ is written as a linear combination $c_1\mathbf{u} + c_2\mathbf{v}$. Then

$$\begin{aligned}\mathbf{y} \cdot \mathbf{w} &= \mathbf{y} \cdot (c_1\mathbf{u} + c_2\mathbf{v}) \\ &= \mathbf{y} \cdot c_1\mathbf{u} + \mathbf{y} \cdot c_2\mathbf{v} \\ &= c_1\mathbf{y} \cdot \mathbf{u} + c_2\mathbf{y} \cdot \mathbf{v} \\ &= c_1(0) + c_2(0).\end{aligned}$$

The last line used the assumption that $\mathbf{y} \cdot \mathbf{u} = \mathbf{y} \cdot \mathbf{v} = 0$.

This proves that $\mathbf{y}$ is orthogonal to every vector in $\text{Span}\{\mathbf{u}, \mathbf{v}\}$.